

Network Security Lab		Semester	VI
Course Code	BCYL606	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	0:0:2:0	SEE Marks	50
Credits	01	Exam Hours	100
Examination type (SEE)	Practical		
Course objectives: • To get Practical exposure of Substitution Techniques • To get Practical exposure on Symmetric Cryptographic Algorithms • To get Practical exposure on Asymmetric Cryptographic Algorithms •			
SL.NO	Experiments (Implement using C/C++/Java Programming Languages)		
1	Implement Caesar Cipher substitution method		
2	Implement Mono alphabetic cipher with the given key and input		
3	Implement Poly alphabetic Cipher		
4	Encrypt the given text using Play fair Cipher with the given key. Also perform the decryption operation		
5	Implement Encryption and Decryption techniques in Hill Cipher		
6	Demonstrate Single and Double Transposition techniques		
7	Implement Simple DES/DES Algorithm		
8	Generate Pseudo random numbers using Linear Congruential method		
9	Generate Pseudo random numbers using Blum Blum Shub Generator		
10	Implement RSA Algorithm		
11	Demonstrate Diffie Hellman Key exchange Algorithm		
12	Implement Fermat’s and Euler’s Theorem (Course Instructor need to explain the theorem before execution as the topic not covered in Theory part)		
Course outcomes: At the end of the course the student will be able to: • Implement Substitution Ciphers • Design the various Transposition Techniques • Demonstrate the working of various Symmetric Cipher algorithms • Demonstrate the working of various asymmetric Cipher algorithms			

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together

Continuous Internal Evaluation (CIE):

CIE marks for the practical course are **50 Marks**.

The split-up of CIE marks for record/ journal and test are in the ratio **60:40**.

- Each experiment is to be evaluated for conduction with an observation sheet and record write-up. Rubrics for the evaluation of the journal/write-up for hardware/software experiments are designed by the faculty who is handling the laboratory session and are made known to students at the beginning of the practical session.
- Record should contain all the specified experiments in the syllabus and each experiment write-up will be evaluated for 10 marks.
- Total marks scored by the students are scaled down to **30 marks** (60% of maximum marks).
- Weightage to be given for neatness and submission of record/write-up on time.
- Department shall conduct a test of 100 marks after the completion of all the experiments listed in the syllabus.
- In a test, test write-up, conduction of experiment, acceptable result, and procedural knowledge will carry a weightage of 60% and the rest 40% for viva-voce.
- The suitable rubrics can be designed to evaluate each student's performance and learning ability.
- The marks scored shall be scaled down to **20 marks** (40% of the maximum marks).

The Sum of scaled-down marks scored in the report write-up/journal and marks of a test is the total CIE marks scored by the student.

Semester End Evaluation (SEE):

- SEE marks for the practical course are 50 Marks.
- The examination schedule and names of examiners are informed to the university before the conduction of the examination. These practical examinations are to be conducted between the schedule mentioned in the academic calendar of the University.
- All laboratory experiments are to be included for practical examination.
- (Rubrics) Breakup of marks and the instructions printed on the cover page of the answer script to be strictly adhered to by the examiners. **OR** based on the course requirement evaluation rubrics shall be decided jointly by examiners.
- Students can pick one question (experiment) from the questions lot prepared by the examiners jointly.

- Evaluation of test write-up/ conduction procedure and result/viva will be conducted jointly by examiners.

General rubrics suggested for SEE are mentioned here, writeup-20%, Conduction procedure and result in -60%, Viva-voce 20% of maximum marks. SEE for practical shall be evaluated for 100 marks and scored marks shall be scaled down to 50 marks (however, based on course type, rubrics shall be decided by the examiners)

Change of experiment is allowed only once and 15% of Marks allotted to the procedure part are to be made zero.

The minimum duration of SEE is 02 hours

Suggested Learning Resources:

- <https://www.bugcrowd.com/glossary/symmetric-encryption-algorithms/>
- <https://www.ibm.com/think/topics/symmetric-encryption>
- <https://www.wikihow.com/Create-Substitution-Ciphers>